ELECTRIC VEHICLE MARKET ANALYSIS

A Data Analytics Project using MySQL, Tableau and Flask

Submitted by: Pushpendra Sirvi | B.Tech CSE, Semester 3 | Medi-Caps University | Academic Year 2026–2027

Abstract

This project demonstrates an educational data analytics workflow using a supplied EV dataset, MySQL, Tableau and Flask. The dataset is synthetic/educational and must not be treated as verified real-world market statistics.

1. Problem Understanding

EV records contain category, manufacturer, battery type, capacity, range, price, production country, customer rating and sales-unit attributes. This project makes these fields easier to explore through SQL and visual analytics.

2. Objectives

Inspect data; design MySQL schema; perform SQL analysis; create seven distinct visualization types; build an interactive dashboard and six-scene story; embed published Tableau views in Flask; test and document actual behavior.

3. Stakeholders

Emily Johnson — market strategy: category/manufacturer comparison and ratings. David Chen — BI: rankings, battery distribution and production geography. Dr. Sarah Williams — research: battery, range, capacity and satisfaction analysis.

4. Dataset audit

Training CSV: 200,000 rows and 37 columns; test CSV: 100,000 rows. Training data includes 15 categories, 5 battery types and 200 manufacturers. Audit found 18,996 missing cells and 1 repeated vehicle ID occurrence(s). Mean range: 324.5 km; mean capacity: 78.2 kWh; mean rating: 4.09/5.

5. Database & SQL

Database ev_market_analysis; table ev_vehicles; intended primary key vehicle_id. SQL scripts include quality checks, category sums, manufacturer ranking, battery summary, average range/capacity, ratings and a category summary view. Clean duplicate IDs and document null handling before import.

6. Tableau visualization types

1) Horizontal bar: sample sales units by category. 2) Treemap: distinct records by battery type. 3) Scatter plot: capacity versus range. 4) Highlight table: rating by manufacturer and battery type. 5) Map: records by manufacturing country. 6) Box plot: range distribution by category. 7) Packed bubbles: records by battery type.

7. Dashboard & Story

Dashboard: KPI cards and filters for model year, manufacturer, category, battery type, manufacturing country and market region. Six story scenes: dataset snapshot, category comparison, manufacturer comparison, battery mix, range/capacity/ratings, and production footprint with recommendations.

8. Flask integration

The Flask site has home, dashboard and story routes. Set TABLEAU_DASHBOARD_URL and TABLEAU_STORY_URL after publishing views to an accessible Tableau server. Check sharing/privacy settings before embedding.

9. Testing

Record actual results for CSV schema, duplicate IDs, missing values, SQL/Tableau reconciliation, filters, responsive layouts, Flask routes and load times. Do not invent performance measurements.

10. Limitations & conclusion

The supplied data is synthetic/educational. Sales sums describe dataset records only. model_year is not a validated sales time series; manufacturing_country is production origin. The project demonstrates a reproducible educational workflow, not real-world

market forecasting.